

Cenomi Chatbot — Status Report

As of: Wednesday, 25 March 2026

Demo target: Sunday, 30 March 2026 (5 days)

1. Where We Are on the Improvisation — What Has Been Implemented

The platform has shipped four full versions since the initial build on 13 March. Every change has been verified at 100% pass rate across the 204-query test suite (smoke + adversarial multi-turn + deep complex scenarios).

v1.0 — 13 March: Core Pipeline (Single Mall)

- **LangGraph pipeline** — 12-node directed graph covering: session load → intent interpretation → scene memory update → playbook resolution → strategy selection → context composition → retrieval decision → fact fetch → response generation → memory update → debug payload emit
- **Intent classification** — three-tier hybrid: short-circuit rule table (< 1ms) → regex rule-based classifier → GPT-4o LLM fallback. Domains: dining, shopping, entertainment, services, navigation, exploration, mall_info, general
- **Scene memory** — accumulates companions, occasion, budget, visit plan, topic lock, active shortlist across every turn; resolves terse follow-ups in context
- **Playbook engine** — scenario playbooks matched and ranked against intent + entity context; tone-mapped outputs (date night vs family day vs gift hunt produce structurally different responses)
- **Anti-hallucination guard** — post-generation validation of all LLM output against canonical mall data before delivery
- **Feedback system** — explicit (thumbs up/down) + implicit (auto-detected corrections); session tuning engine applies signals in real time during the session
- **React testing console** — full debug inspector (intent, playbook, entities, retrieval, node trace, latencies), session export
- **Mall loaded:** Al Nakheel Plaza, Buraidah (mall 28) — 94 brands, 1 cinema, 6 movies, 10 services

v1.1 — 17 March: Multi-Mall Support

- Runtime now holds a registry of mall contexts instead of a singleton
- Mall of Arabia, Jeddah (mall 13) added — full data pipeline run (canonical, semantic, playbook, tenant config)
- All pipeline nodes updated to use `mall_id`-scoped context
- Frontend mall selector updated; health endpoint returns `mall_ids` list

v1.2 — 17 March: Cross-Mall Brand Search

- New `cross_mall` intent domain — fires for queries like "Does Mall of Arabia also have Nike?" or "Which Cenomi mall has Starbucks?"
- `search_brand_across_malls()` merges canonical data across all loaded malls, sorts home mall first
- Dedicated `_build_cross_mall_response()` with structured `AT YOUR CURRENT MALL / AT OTHER CENOMI MALLS` prompt blocks
- Hallucination guard extended to validate against a merged canonical index from all loaded malls

v1.3 — 23 March: Infrastructure & Streaming

- **Redis session persistence** — sessions survive backend restarts and are shareable across workers; `RedisSessionStore` with configurable TTL; in-memory fallback when Redis is absent
- **LangGraph turn checkpointing** — every graph invocation snapshot-persisted to Redis (`BACKEND_ENABLE_CHECKPOINTER=true`); each turn is independently replayable for debugging and regression testing
- **Async quality evaluator** — LLM-as-judge evaluates every response on `intent_alignment`, `constraint_adherence`, `honesty`, and `conciseness` (0–10); runs in a background task, never blocks delivery
- **SSE token streaming** — `POST /api/chat/stream` streams response token-by-token; React UI renders live blinking cursor; session persistence and quality evaluation run post-stream without delaying the first token
- **Docker / Docker Compose** — single-command deployment; multi-stage Dockerfile under 300 MB; Redis + backend services with healthcheck; `backend/data` volume mount for persistence

v1.4 — 24 March: Semantic Retrieval + Multi-Mall Scale

- **Three-layer hybrid retrieval:**
- **Layer 1 — Category rules:** deterministic lookup, 100% precision, zero LLM cost (~0ms)
- **Layer 2 — Canonical name search:** substring + word-boundary match for brand-specific queries
- **Layer 3 — Vector similarity search:** `VectorStoreService` (Chroma + `text-embedding-3-small` + Redis result cache); falls back to tag-overlap matching when Chroma is unavailable
- **VectorStoreService** — Chroma PersistentClient (HNSW cosine, 1536-dim); Redis-backed result cache (TTL configurable); per-mall collections with zero cross-contamination
- **Multi-mall LRU cache (three-tier resource design):**
- Tier 1 (Python RAM) — LRU cache of active `MallContextLoader` objects (default capacity 5); eviction frees RAM
- Tier 2 (Redis) — serialized mall context JSON with TTL; restoring from Redis is ~50–100× faster than disk
- Tier 3 (Disk) — cold load from canonical/semantic/playbook JSON; writes back to Redis on first load
- **Experience layer** (`experience_resolver.py`) — selects response composition mode before the LLM is called: `micro_itinerary`, `curated_shortlist`, `filtered_factual_list`, `guided_plan`, `structured_overview`

- **Pronoun reference fix** — queries like "anything she would like" now correctly route to `guided_recommendation` instead of treating "she" as a brand name
- **Comprehensive test runner** (`run_all_tests.py`) — runs all three test suites against the live Docker backend; generates Markdown + JSON reports in `test-results/`

v1.4.1 — 24 March: Contextual Query Augmentation

- `build_scene_prefix(scene)` prepends `SceneMemory` signals (occasion, companions, visit_type, budget) to the vector query before embedding
- The same bare query from different visitor contexts now produces different embedding vectors:
- Anniversary couple → embed `"anniversary partner date something for dinner"`
- Family with kids → embed `"kids family outing something for dinner"`
- Fresh session → embed `"something for dinner"` (unchanged — no regression risk)
- **Zero regressions: 204/204 test cases pass (100%)**

2. What Will Be Included Before the Demo (30.03.26)

The demo is in 5 days. Based on the roadmap priorities and effort estimates documented in `docs/COMPETITIVE-ANALYSIS-AI-CORE-RETRIEVAL.md` and `docs/AI-FINDR-COMPARISON.md` , the following items are **realistic to complete before 30 March**:

Retrieval Improvements (can be done before demo)

Item	Effort	Impact	Status
BM25 entity name index + RRF fusion	1–2 days	High — closes the exact brand name miss gap vs vector-only search	Pending
Cross-encoder re-ranker (top-20 → top-5, <code>ms-marco-MiniLM-L-6-v2</code> , ~25MB, CPU-runable)	< 1 day	Medium — improves ranking for ambiguous queries by ~15–25%	Pending
Vector fallback in factual flow (<code>fetch_exact_facts</code> falls back to vector when canonical lookup returns empty)	< 1 day	Medium — surfaces answers for service/facility queries that miss keyword lookup	Pending

Demo Preparation (should be done before demo)

Item	Notes
README update	Current README describes old 12-node single-flow architecture; live system is a 16-node dual-flow graph. Should reflect actual architecture and API routes.

Item	Notes
Delta-ingest trigger (mtime-based, background)	Medium effort (2–3 days) — borderline for the demo window

Items NOT Realistic Before 30.03.26

Item	Reason
Arabic multilingual support	High effort — language detection, Arabic prompt variants, regional dialect handling; not achievable in 5 days
Analytics dashboard	Requires database migration + new frontend; 1+ week of work
WhatsApp integration	Requires WhatsApp Business API registration + integration work
Visual / rich responses (cards, maps, CTAs)	Requires new response schema + frontend rendering overhaul
LangSmith production tracing	Config exists but never wired into any code path

3. How Close We Are with AI Findr

Overall feature parity: ~60–65%

The key insight is that the gap is **not in AI quality** — it is in the delivery and operationalization layers around the AI core.

Conversational AI core		~87% parity (Cenomi leads on depth)
Retrieval quality		~65% parity (three-layer hybrid + scene augme
Omnichannel delivery		~20% parity (SSE streaming + text API; no Wha
Visual / rich responses		~10% parity (plain text only)
Analytics & insights		~20% parity (JSON files vs dashboard)
Non-technical config		~10% parity (code-only config)
Multilingual (Arabic)		0% parity (not implemented)

Where Cenomi Wins Outright

- **Conversational depth** — multi-turn scene memory, visit plan tracking, companion/occasion/budget accumulation across turns; AI Findr is query-response only
- **Dual-flow routing** — 7 weighted routing rules decide per-query whether to use a 9-node concierge flow or a 7-node factual flow; AI Findr is a single flow
- **Hallucination guard** — explicit post-generation validation against structured canonical data; AI Findr claims grounding but details are opaque and unpublished

- **Real-time session adaptation** — implicit feedback detection + `session_tuning_engine` adjusts behavior mid-session; AI Findr only improves offline at the platform level
- **Playbook scenario engine** — dynamic matching + tone mapping per scenario; AI Findr's "experience builder" is a static config UI
- **Full data ownership and Cenomi-specific tuning** — Gulf regional occasions, local brands, Saudi cultural context embedded natively
- **Debug observability** — full node trace, per-step latency, retrieval decisions, scene state snapshot per turn; AI Findr is a black box
- **Data residency** — fully self-hosted; AI Findr requires Enterprise tier for on-premises

Where AI Findr Wins Outright

- **Omnichannel** — WhatsApp, kiosk, digital signage, voice, mobile app; Cenomi is web-only
- **Visual responses** — interactive maps, store cards with images, event banners, CTA buttons
- **Multilingual / Arabic** — auto-detect with regional tone calibration; Cenomi has zero Arabic support
- **Analytics dashboard** — real-time top queries, trends, content gaps, per-channel performance
- **No-code config** — marketing teams can change tone, rules, and playbooks without engineering
- **Lead capture** — unanswered queries become warm leads; Cenomi gives a best-effort answer or acknowledges the gap
- **Tenant monetization** — promoted listings with tracked impressions, clicks, conversions
- **Proven production results** — 87.3% first-contact resolution, +20% conversion, +39 NPS at Jockey Plaza and Multiplaza Bogotá
- **Fast deployment** — working prototype in under one week; no data loading required

4. What AI Findr Doesn't Have That We Have (and Are Building)

This is where the Cenomi chatbot's architectural investment creates durable differentiation.

Already Implemented (AI Findr Cannot Match)

Capability	What It Does	Why AI Findr Can't Match
SceneMemory	Tracks companions, occasion, budget, topic lock, visit plan, shopping task, constraints across every turn	AI Findr is query-response — no multi-turn planning state published
Dual-flow LangGraph pipeline	Routes each query through either concierge flow (planning, curation) or factual flow (hours, availability) based on 7 weighted rules after intent interpretation	Single flow — no concierge vs factual separation
Post-generation hallucination guard	Every LLM output validated against canonical mall data (store names, hours, movies, offers) before delivery	Claims grounding but no post-generation validation

Capability	What It Does	Why AI Findr Can't Match
		step is published
Real-time within-session behavioral adaptation	Explicit (thumbs) + implicit (auto-detected corrections) feedback tuned mid-session via <code>session_tuning_engine</code>	Only platform-level self-improvement over time; no real-time session adaptation
Playbook scenario engine with tone mapping	"Romantic dinner" playbook produces different entity ranking and tone than "family outing" over identical data	Static experience builder UI — not dynamically matched to conversation context
Cross-mall canonical search with guard	"Is Zara in any Cenomi mall?" searches all loaded mall contexts; answer validated against merged canonical index	Per-mall configuration — no cross-property canonical search
LangGraph turn checkpointing	Every graph turn snapshot-persisted and independently replayable for debugging/regression testing	Black box — errors require a vendor support ticket
Full debug observability API	Node trace, per-step latency, detected intent, retrieved entities, scene state snapshot per turn	No debug API published
Async quality evaluator (LLM-as-judge)	Scores <code>intent_alignment</code> , <code>constraint_adherence</code> , <code>honesty</code> , <code>conciseness</code> on every response without blocking delivery	No equivalent published
Knowledge gap analyzer	Surfaces specific query topics the bot failed to answer, feeding content team updates	Not published
Scene-augmented vector query	<code>SceneMemory</code> context (occasion, companions, visit_type, budget) prepended to query before embedding — different sessions get different retrieval results from identical bare queries	Static indexing — the same query always returns the same results
Three-tier LRU mall context cache	Serves 20 malls with bounded RAM using RAM → Redis → Disk tiering	No published multi-mall context management architecture
Full data ownership	Prompts, retrieval rules, playbooks, entity data all owned by Cenomi — Gulf/Saudi-specific context embedded natively	Generic SaaS platform — cannot match Cenomi/Gulf specificity by default
Self-hosting with no data residency dependency	All visitor query data stays on Cenomi infrastructure by default	Data leaves infrastructure on standard tiers; on-premises is Enterprise-only

Capability	What It Does	Why AI Findr Can't Match
LLM cost efficiency	Rule fast-path handles ~80% of queries with zero LLM cost (< 1ms); GPT-4o only for ambiguous queries	All queries incur full LLM cost

Being Built (Pre-Demo and Post-Demo Roadmap)

Capability	Roadmap	What It Will Add
BM25 + RRF hybrid retrieval	Pre-demo (P2 retrieval roadmap)	Closes the failure mode where exact brand names in unusual embedding positions miss vector recall; fuses keyword precision with semantic recall
Cross-encoder re-ranker	Pre-demo (P3 retrieval roadmap)	Post-vector step reranking top-20 candidates by true relevance, improving ambiguous query quality by ~15–25%
Vector fallback in factual flow	Pre-demo (P4 retrieval roadmap)	Factual queries that miss keyword lookup fall back to vector search — fewer "not found" responses
Arabic multilingual support	Post-demo (Priority 2 in gap roadmap)	Language detection, Arabic prompt variants, bilingual vector indexing, regional dialect handling
Analytics dashboard	Post-demo (Priority 3)	Connects existing backend analytics infrastructure (feedback events, evaluator results, knowledge gaps) to a UI business teams can act on
WhatsApp integration	Post-demo (Priority 4)	Near-universal Gulf channel — wraps existing <code>/api/chat</code> endpoint, maps WhatsApp sender IDs to sessions
Rich / visual responses	Post-demo (Priority 5)	Store cards, map links, CTA buttons alongside plain text — higher conversion

Summary

Question	Short Answer
Improvisation status	5 shipped versions (v1.0–v1.4.1) since 13 March. Platform is at 204/204 (100%) test pass rate. Full hybrid retrieval, streaming, Docker, multi-mall, scene-augmented vectors all live.
Before demo (30.03.26)	BM25 hybrid retrieval, cross-encoder re-ranker, and vector fallback in factual flow are the 3 realistic additions. README update is a must. Arabic, analytics, WhatsApp, and visual responses are post-demo.
Proximity to AI Findr	~60–65% overall feature parity. ~87% on conversational AI core (Cenomi leads). ~20–0% on delivery/operational layer (AI Findr leads).

Question	Short Answer
What only we have	Scene memory, multi-turn planning, dual-flow routing, post-generation hallucination guard, real-time session adaptation, playbook engine, scene-augmented vector retrieval, full data ownership + Gulf tuning, LangGraph observability, and LLM cost efficiency via rule fast-path.